

Positive Topology and Feasible Refinement: Forcing Matrices, Positivity, and Information

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Abstract

Starting from a point–basic–open forcing relation $x \Vdash a$ between a set of points/models X and basic opens/generators S , we obtain the canonical Galois adjunction $\text{ext} \dashv \square$ between $\mathcal{P}(S)$ and $\mathcal{P}(X)$ encoding the “open/locale” (universal, cover/refinement) content [5]. Less standard, but equally canonical, is a second adjunction $\diamond \dashv \text{rest}$ encoding the “closed/positive” (existential, probe/meet) content; this yields a direct origin for Sambin-style positivity $a \ltimes U$ and “closed without complements” as fixed points of the contractive operator $\mathcal{J} = \diamond \circ \text{rest}$ [1, 6]. We isolate a reconstruction principle: the forcing matrix is recovered uniquely from either residual pair via singleton tests.

When points are given, positivity can be induced from forcing in this way; in the formal pointfree approach, positivity is instead taken as primitive and compatibility with cover is axiomatic. This distinction supports two operational readings [1, 4]: (i) an information-theoretic view, where covers express refinement of partial information and positivity expresses witnessed feasibility, and (ii) a game view, where a positive witness supplies a strategy through successive cover refinements.

The final part of the survey is deliberately programmatic. We sketch how resource annotations may refine ordinary forcing, how budget-indexed covers and positivity may then be studied, and how enriched variants suggest resource-aware semantics. We do *not* claim here a final resource logic: the interaction among budgeted forcing, cover, positivity, and logical connectives is deferred to a subsequent article [7]. Examples from medical diagnosis, legal reasoning, and AI systems illustrate the intended direction.

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1 Introduction: Why Positive Topology?

Before diving into the formal machinery, we motivate the key ideas through a concrete scenario.

1.1 A motivating example: Medical diagnosis

Consider the task of diagnosing a patient based on observed symptoms.

- X = patients (the “points” or “states of the world”)
- S = symptoms, tests, conditions (the “observables” or “basic opens”)
- $x \Vdash a$ means “patient x exhibits symptom/condition a ”

A doctor observes that a patient has fever, cough, and fatigue. What can we infer?

The cover relation: semantic entailment. Certain conditions *imply* certain symptoms. For instance:

$$\text{pneumonia} \triangleleft \{\text{fever, cough, chest_pain, infiltrates}\}$$

This means: every patient with pneumonia exhibits at least one of fever, cough, chest pain, or lung infiltrates. More generally, $a \triangleleft U$ says: *whenever a holds, something in U holds*.

This is the **universal/refinement** side: covers encode what implies what.

The positivity relation: witnessed consistency. Now consider the question: is the symptom cluster {fever, cough, fatigue} a *realizable* profile? That is, does there exist a patient whose *complete* symptom profile is contained in this set?

We write:

$$\text{fever} \ltimes \{\text{fever, cough, fatigue}\}$$

to mean: there exists a patient with fever whose entire observable profile is contained in the displayed set.

This patient is a **witness** — the symptom cluster is realizable, not a contradiction.

Contrast with:

$$\text{fever} \ltimes \{\text{hypothermia}\}$$

This is *false*: no patient has both fever and only hypothermia. The combination is inconsistent.

This is the **existential/positivity** side: positivity encodes what is consistent, what has witnesses.

Why both matter.

- **Covers alone** tell us what implies what, but not what’s realizable.
- **Positivity alone** tells us what’s consistent, but not what follows from what.
- **Together** they give us: grounded inference (every conclusion has a witness) with semantic structure (implications are tracked).

The compatibility axiom. The key interaction between covers and positivity is:

$$(a \triangleleft U) \wedge (a \ltimes V) \implies \exists u \in U. u \ltimes V$$

In medical terms: if pneumonia implies {fever, cough, infiltrates}, and there exists a pneumonia patient confined to profile V , then there exists a fever patient (or cough patient, or infiltrates patient) also confined to V .

Refinement preserves witnesses. This is the soundness condition that makes the framework coherent.

1.2 The general pattern

The medical example instantiates a universal pattern:

Domain	Points X	Observables S
Medicine	Patients	Symptoms, tests, conditions
Law	Cases	Evidence, claims, precedents
Programming	Program states	Properties, assertions, tests
Finance	Portfolios	Risk factors, exposures
Science	Hypotheses	Experiments, observations
AI agents	Traces/executions	Constraints, evaluations

This operational reading follows the constructive-positivity tradition initiated by Sambin [1] and formalized via inductively generated formal topologies [6]. It abstracts the familiar locale/sheaf perspective of Mac Lane–Moerdijk [5] down to the level of the forcing matrix we use in code.

In each domain:

- $a \triangleleft U$ encodes *semantic entailment*: a implies something in U
- $a \times U$ encodes *witnessed consistency*: there exists x with $x \Vdash a$ and $\Diamond x \subseteq U$
- The forcing $x \Vdash a$ connects the abstract observables to concrete states

The rest of this paper develops these ideas formally through the forcing/adjunction/positivity construction. The resource-aware part is intentionally an *esquisse d'un programme*: it records a plausible direction and isolates questions whose full treatment is left to the subsequent article [7].

2 The starting datum: a forcing/incidence relation

Definition 2.1 (Forcing table). Fix sets X (points/models) and S (basic opens/generators). A *forcing relation* is a subset $R \subseteq X \times S$ written

$$x \Vdash a \quad \text{for } (x, a) \in R.$$

Remark 2.2 (Interpretation). Think of $a \in S$ as a basic open (or observable, or test) and $x \in X$ as a point (or state, or model). Then $x \Vdash a$ reads: “ x lies in the basic open a ” or “ x satisfies observable a ”. No topology is assumed at the outset: we start from the incidence matrix.

Example 2.3 (Medical diagnosis). Let X be a set of patients and S a set of symptoms/conditions. Then $x \Vdash a$ means “patient x exhibits symptom a ”. The forcing relation is the patient-symptom incidence matrix.

Example 2.4 (Program verification). Let X be program states and S be assertions/properties. Then $x \Vdash a$ means “state x satisfies property a ”.

Remark 2.5 (Boolean matrix viewpoint). The relation $R \subseteq X \times S$ is a Boolean-valued matrix $R : X \times S \rightarrow \{0, 1\}$. Much of what follows is “linear algebra over $\{0, 1\}$ ”: joins are unions, meets are intersections. This matrix viewpoint also suggests the enriched, resource-sensitive direction sketched in Section 9.

3 The first adjunction: opens via $\text{ext} \dashv \Box$

3.1 Direct image and the universal residual

Definition 3.1 (Extension and interior-kernel). Define monotone maps

$$\begin{aligned} \text{ext} : \mathcal{P}(S) &\rightarrow \mathcal{P}(X), & \text{ext } U &= \{x \in X : \exists a \in U, x \Vdash a\}, \\ \Box : \mathcal{P}(X) &\rightarrow \mathcal{P}(S), & \Box E &= \{a \in S : \forall x \in X, (x \Vdash a \Rightarrow x \in E)\}. \end{aligned}$$

Example 3.2 (Medical: Extension and Interior). Continuing Example 2.3:

- $\text{ext} \{\text{fever, cough}\} = \text{patients who have fever OR cough (or both)}$
- $\Box E$ where $E = \text{“patients over 65”}$ gives: symptoms that *only* occur in patients over 65

Proposition 3.3 (Galois adjunction I). For all $U \subseteq S$ and $E \subseteq X$,

$$\text{ext } U \subseteq E \iff U \subseteq \Box E.$$

Equivalently, $\text{ext} \dashv \Box$.

Proof. $(\Rightarrow)$ If $a \in U$, then $\text{ext } \{a\} \subseteq \text{ext } U \subseteq E$, hence $a \in \Box E$. $(\Leftarrow)$ If $x \in \text{ext } U$, choose $a \in U$ with $x \Vdash a$. Since $a \in \Box E$, we have $x \in E$. $\square$

Remark 3.4 (Universal nature). $\Box E$ is defined by a $\forall$ -condition; this is the “universal/refinement” side. This is exactly why the cover relation induced by forcing is a universal statement.

3.2 Cover and the pointfree (formal-cover) side

Definition 3.5 (Induced preorder and cover). Define a preorder on S by

$$a \leq b \iff \forall x, (x \Vdash a \Rightarrow x \Vdash b),$$

and define a cover relation $\triangleleft$ by

$$a \triangleleft U \iff \forall x, (x \Vdash a \Rightarrow \exists b \in U, x \Vdash b).$$

Example 3.6 (Medical: Cover as symptom entailment). In the medical setting, $a \triangleleft U$ means: every patient with condition a exhibits at least one symptom in U . For instance:

$$\begin{aligned} \text{flu} &\triangleleft \{\text{fever, cough, fatigue, myalgia}\} \\ \text{bacterial_infection} &\triangleleft \{\text{fever, elevated_WBC}\} \end{aligned}$$

These are *diagnostic implications*: the disease implies certain observable symptoms.

Remark 3.7 (Extensional reading). Equivalently:

$$a \leq b \iff \text{ext } \{a\} \subseteq \text{ext } \{b\}, \quad a \triangleleft U \iff \text{ext } \{a\} \subseteq \text{ext } U.$$

So $\triangleleft$ is literally “inclusion in a union”.

3.3 Extensional topology on points

Definition 3.8 (Generated opens on X). A subset $E \subseteq X$ is *extensional open* (generated by S) if

$$E = \text{ext} \square E.$$

Remark 3.9. The operator $\text{int} := \text{ext} \circ \square$ is an *interior* operator on $\mathcal{P}(X)$: it extracts the largest extensional open contained in a given subset. (Dually, $\mathcal{A} := \square \circ \text{ext}$ is the *saturation* operator, a closure/nucleus-like operator on $\mathcal{P}(S)$.) Extensional opens are exactly the fixed points of int : “those subsets describable by generators”.

Definition 3.10 (Extensional equivalence of points). Define $x \sim y$ if the induced neighborhood sets agree:

$$x \sim y \quad :\Longleftrightarrow \quad \forall a \in S, (x \Vdash a \Leftrightarrow y \Vdash a).$$

Example 3.11 (Medical: Indistinguishable patients). Two patients $x \sim y$ are *symptomatically indistinguishable*: they exhibit exactly the same symptoms. From a diagnostic standpoint, they are equivalent — the same inferences apply to both.

Remark 3.12. Pointfree semantics does not distinguish points inside the same $\sim$ -class. This is the basic “quotient-to-locale” phenomenon: locale data sees neighborhood behavior, not point multiplicity.

4 The second adjunction: positivity via $\Diamond \dashv \text{rest}$

The locale/cover side is universal; “closed without complements” needs an existential probe: “does a basic open *hit* a closed thing?”. This is the conceptual origin of Sambin-style positivity [1, 3, 4].

4.1 Neighborhood sets and constraint-selected points

Definition 4.1 (Neighborhood set). For $x \in X$, its neighborhood set (or *observable profile*) is

$$\Diamond x := \{a \in S : x \Vdash a\} \subseteq S.$$

Example 4.2 (Medical: Symptom profile). For a patient x , the neighborhood set $\Diamond x$ is their *complete symptom profile*: the set of all symptoms/conditions they exhibit.

Definition 4.3 (Selection by a constraint on neighborhoods). Define

$$\text{rest} : \mathcal{P}(S) \rightarrow \mathcal{P}(X), \quad \text{rest } U := \{x \in X : \Diamond x \subseteq U\}.$$

Thus $\text{rest } U$ consists of points all of whose basic neighborhoods lie in U .

Example 4.4 (Medical: Patients confined to a symptom cluster). $\text{rest } \{\text{fever}, \text{cough}, \text{fatigue}\} =$ patients whose *entire* symptom profile is contained in $\{\text{fever}, \text{cough}, \text{fatigue}\}$.

These are patients who have *only* fever, cough, and/or fatigue — no other symptoms.

4.2 The existential “hit” map

Definition 4.5 ($\diamond$ -Hit). Define

$$\diamond : \mathcal{P}(X) \rightarrow \mathcal{P}(S), \quad \diamond D := \{a \in S : \exists x \in D, x \Vdash a\}, \quad \text{that is, } a \varepsilon \diamond D \leftrightarrow \text{ext } a \not\Downarrow D.$$

In words, a hits the set D if a has a witness inside D .

Example 4.6 (Medical: Symptoms witnessed in a cohort). If $D = \text{patients in a clinical trial}$, then $\diamond D = \text{symptoms exhibited by at least one patient in the trial}$.

Proposition 4.7 (Galois adjunction II). *For all $D \subseteq X$ and $U \subseteq S$,*

$$\diamond D \subseteq U \iff D \subseteq \text{rest } U.$$

Equivalently, $\diamond \dashv \text{rest}$.

Proof. ($\Rightarrow$) Let $x \in D$. If $a \in \diamond x$ then $a \in \diamond D \subseteq U$. Hence $\diamond x \subseteq U$ and $x \in \text{rest } U$.

($\Leftarrow$) If $a \in \diamond D$, choose $x \in D$ with $x \Vdash a$. Since $x \in \text{rest } U$ we have $a \in \diamond x \subseteq U$. $\square$

Remark 4.8 (Meaning). $\diamond D$ records which generators are *witnessed* in D . $\text{rest } U$ picks points whose *entire neighborhood behavior* is constrained by U . This is the “closed-without-complements” move: U acts as a *code* for a closed-like object by constraining neighborhoods, not by negation.

5 Positivity and closed sets without complements

5.1 The positivity operator $\mathcal{J} = \diamond \circ \text{rest}$

Definition 5.1 (Positivity interior on codes). Define

$$\mathcal{J} : \mathcal{P}(S) \rightarrow \mathcal{P}(S), \quad \mathcal{J}U := \diamond \text{rest } U.$$

Equivalently,

$$a \varepsilon \mathcal{J}U \leftrightarrow \exists x \in X (x \Vdash a \wedge \diamond x \subseteq U) \leftrightarrow \text{ext } a \not\Downarrow \text{rest } U.$$

Definition 5.2 (Positivity relation). Define $\ltimes \subseteq S \times \mathcal{P}(S)$ by

$$a \ltimes U \iff a \in \mathcal{J}U.$$

Remark 5.3 (One-line intuition). $a \ltimes U$ means: there exists a point x in basic open a such that all basic neighborhoods of x lie in U . So U behaves like a “closed neighborhood specification” and a *hits* it.

Example 5.4 (Medical: Positivity as witnessed symptom clusters). The statement

$$\text{fever} \ltimes \{\text{fever, cough, fatigue}\}$$

means that there exists a patient who has fever and whose *complete* symptom profile is contained in that set.

This patient is a **witness** that fever is consistent with the symptom cluster.

Contrast: $\text{fever} \ltimes \{\text{hypothermia}\}$ is *false* because no patient has both fever and only hypothermia — the combination is contradictory.

Positivity detects **realizability**: which symptom combinations actually occur in patients.

Proposition 5.5 (A point forces positivity). *Assume a point $x \in X$ and a generator $a \in S$ with $x \Vdash a$. Then a is positive in the unary sense, i.e. $a \ltimes S$. More generally, if $U \subseteq S$ satisfies $\Diamond x \subseteq U$, then $a \ltimes U$.*

Proof. If $\Diamond x \subseteq U$ then $x \in \text{rest } U$ by definition, hence $a \in \Diamond \text{rest } U = \mathcal{J}U$, i.e. $a \ltimes U$. Taking $U = S$ gives $a \ltimes S$. $\square$

5.2 Elementary properties

Proposition 5.6 (Contractive and monotone). *For all $U, V \subseteq S$:*

(a) $\mathcal{J}U \subseteq U$.

(b) If $U \subseteq V$ then $\mathcal{J}U \subseteq \mathcal{J}V$.

Proof. (a) If $a \in \mathcal{J}U$, choose x with $x \Vdash a$ and $\Diamond x \subseteq U$. Then $a \in \Diamond x \subseteq U$.

(b) If $U \subseteq V$ then $\text{rest } U \subseteq \text{rest } V$, hence $\Diamond \text{rest } U \subseteq \Diamond \text{rest } V$. $\square$

Proposition 5.7 (Idempotence). $\mathcal{J} = \Diamond \circ \text{rest}$ is idempotent: $\mathcal{J}\mathcal{J}U = \mathcal{J}U$.

Proof. Since $\mathcal{J} \leq \text{id}$, we have $\mathcal{J}\mathcal{J}U \subseteq \mathcal{J}U$. For the reverse, $\text{id} \leq \text{rest} \circ \Diamond$ implies $\text{rest } U \subseteq \text{rest } \mathcal{J}U$; applying $\Diamond$ yields $\mathcal{J}U \subseteq \mathcal{J}\mathcal{J}U$. $\square$

Definition 5.8 (Formal closed (via positivity)). Call $U \subseteq S$ *formal closed* if $\mathcal{J}U = U$.

Example 5.9 (Medical: Complete symptom profiles). Because $\mathcal{J}U \subseteq U$ holds for every U , the fixed-point condition $\mathcal{J}U = U$ amounts to the converse inclusion $U \subseteq \mathcal{J}U$. Thus a set $U \subseteq S$ is formal closed iff, for every symptom $a \in U$, there exists a patient x such that

$$x \Vdash a \quad \text{and} \quad \Diamond x \subseteq U.$$

In other words, every observable admitted by the closed code U is actually realized by a witness whose complete observable profile stays inside U . This positive formulation is the useful constructive content of closedness here.

5.3 The compatibility axiom

The key interaction between covers and positivity is captured by:

Proposition 5.10 (Compatibility). *If $\mathcal{A}, \mathcal{J}$ are defined as above from $\Vdash$, then for all $U, V \subseteq S$:*

$$\mathcal{A}U \mathbin{\mathcal{J}} \mathcal{J}V \rightarrow U \mathbin{\mathcal{J}} \mathcal{J}V.$$

Proof. Suppose $\mathcal{A}U \mathbin{\mathcal{J}} \mathcal{J}V$, that is, there exists a with $a \in \mathcal{A}U$ and $a \ltimes V$. By positivity ($a \ltimes V$), there exists x with $x \Vdash a$ and $\Diamond x \subseteq V$. By $a \in \mathcal{A}U$, i.e. $\text{ext } a \subseteq \text{ext } U$, since $x \Vdash a$ there exists $u \in U$ with $x \Vdash u$. Then $u \in \Diamond x \subseteq V$, so $u \ltimes V$; hence $U \mathbin{\mathcal{J}} \mathcal{J}V$. $\square$

Compatibility is provable when cover and positivity are defined via points. In a formal approach, it is assumed as an axiom.

Remark 5.11 (Why compatibility matters). Compatibility is the **soundness condition** for refinement: if you refine a hypothesis (via covers), you don't lose your witnesses (positivity). This is what makes the formal framework coherent for inference.

Example 5.12 (Medical: Refinement preserves witnesses). Suppose

$$\text{pneumonia} \triangleleft \{\text{fever, cough, infiltrates}\},$$

and there exists a pneumonia patient confined to profile V .

Then there exists a fever patient (or cough patient, or infiltrates patient) also confined to V .

The diagnostic refinement from “pneumonia” to its symptoms preserves the witness.

6 Information content and refinement games

The formal relations $\triangleleft$ and $\bowtie$ admit an interpretation closer to computation than to classical topology. Think of a generator $a \in S$ as a *partial information state*. A cover $a \triangleleft U$ says that to *settle* the information in a one must proceed to a feasible refinement and end up in *one of* the subcases in U .

Positivity adds one crucial twist: existence is not classical negation of emptiness, but a *witnessed* possibility of continuing the refinement process.

6.1 The refinement game $G(a, U)$

Fix $a \in S$ and a set $U \subseteq S$ of “acceptable outcomes”. Define an infinite two-player game $G(a, U)$:

- The current position is a generator $b \in S$ (initially $b = a$).
- *Refiner* (“Devil”) chooses a cover $b \triangleleft V$.
- *Witness* (“God”) responds by choosing some $v \in V$.
- The next position is $b := v$.
- Witness wins if she can play forever while keeping every chosen position inside U .

Proposition 6.1 (Positivity as a winning condition). *If $a \bowtie U$, then Witness has a winning strategy in $G(a, U)$.*

Proof. By $a \bowtie U$ there exists $x \in X$ with $x \Vdash a$ and $\Diamond x \subseteq U$. Whenever Refiner plays $b \triangleleft V$ with $x \Vdash b$, there exists $v \in V$ with $x \Vdash v$. Witness chooses such v . Since $v \in \Diamond x \subseteq U$, the position stays in U . $\square$

Medical diagnosis as a refinement game. The game $G(\text{fever}, U)$ where U is a diagnostic profile:

- Refiner (skeptic): “You say fever? What kind? Bacterial? Viral?”
- Witness (diagnostician): “Bacterial fever” (chooses refinement in U)
- Refiner: “What symptoms? Give me a cover.”
- Witness: “Elevated WBC” (still in U)

A real patient x witnessing positivity supplies Witness with a strategy that survives every refinement challenge.

Legal reasoning as a refinement game. Let X = legal cases, S = claims/evidence/precedents. The game: opposing counsel (Refiner) demands finer distinctions; the advocate (Witness) must find precedents. Positivity = having a real case to cite at every challenge.

7 Two adjunctions and four operators

The same forcing table gives two adjunctions, hence four operators:

$$\mathcal{P}(S) \overset{\text{ext}}{\underset{\square}{\rightleftarrows}} \mathcal{P}(X) \quad (\text{ext} \dashv \square), \quad \mathcal{P}(X) \overset{\diamond}{\underset{\text{rest}}{\rightleftarrows}} \mathcal{P}(S) \quad (\diamond \dashv \text{rest}).$$

Operator	Type	Meaning
ext	$\mathcal{P}(S) \rightarrow \mathcal{P}(X)$	Points satisfying some observable in U
$\square$	$\mathcal{P}(X) \rightarrow \mathcal{P}(S)$	Observables universally valid on E
$\diamond$	$\mathcal{P}(X) \rightarrow \mathcal{P}(S)$	Observables witnessed in D
rest	$\mathcal{P}(S) \rightarrow \mathcal{P}(X)$	Points confined to profile U
$\mathcal{A} = \square \circ \text{ext}$	$\mathcal{P}(S) \rightarrow \mathcal{P}(S)$	Saturation (formal opens)
int = ext $\circ \square$	$\mathcal{P}(X) \rightarrow \mathcal{P}(X)$	Interior (extensional opens)
$\mathcal{J} = \diamond \circ \text{rest}$	$\mathcal{P}(S) \rightarrow \mathcal{P}(S)$	Reduction (formal closed codes)
cl = rest $\circ \diamond$	$\mathcal{P}(X) \rightarrow \mathcal{P}(X)$	Closure (on points)

Remark 7.1 (Two sides of the same coin). Both adjunctions arise from the same forcing relation, but govern different semantics:

- $\text{ext} \dashv \square$ = universal refinement / locale / covers
- $\diamond \dashv \text{rest}$ = existential probes / positivity / closed-without-complements

8 Reconstruction of forcing from residuals

A central observation (cf. Ciraulo–Sambin [2]) is that the forcing table is recoverable from either left adjoint, since left adjoints preserve joins and are determined by their action on singletons.

Theorem 8.1 (Singleton reconstruction). *If $F = \text{ext}$ arises from $\Vdash$, then $\Vdash$ is uniquely determined by:*

$$x \Vdash a \iff x \in \text{ext} \{a\}.$$

Similarly for $\diamond$:

$$x \Vdash a \iff a \in \diamond \{x\}.$$

Proof. Since ext preserves unions, $\text{ext} U = \bigcup_{a \in U} \text{ext} \{a\}$. Thus $x \in \text{ext} U$ iff $\exists a \in U$ with $x \in \text{ext} \{a\}$. $\square$

Remark 8.2 (What is *not* reconstructible). The composites $j = \square \circ \text{ext}$ or $\mathcal{J} = \diamond \circ \text{rest}$ generally lose witness-level information. The residual *pair* determines forcing; a single composite does not.

9 Feasible refinement and resources: a programmatic sketch

The preceding sections are the mathematically established core of this survey. We now turn to resources. This section is intentionally exploratory: its purpose is to identify a clean baseline and the questions that must be settled in a fuller theory of resource-bounded Positive Topology [7].

9.1 Truth first, cost second: a minimal baseline

A resource annotation should refine ordinary forcing rather than replace truth by affordability. We therefore retain the Boolean forcing relation $x \Vdash a$ and annotate verified incidences with a cost.

Definition 9.1 (Budget preorder). A *budget preorder* is a preorder $(B, \preceq)$. We read $b \preceq b'$ as saying that anything feasible within budget b is also feasible within the more permissive budget b' .

Definition 9.2 (Cost annotation and budgeted forcing). Let

$$c : \{(x, a) \in X \times S : x \Vdash a\} \longrightarrow B$$

assign a verification cost to each true incidence. For $b \in B$ define

$$x \Vdash_b a \iff x \Vdash a \text{ and } c(x, a) \preceq b.$$

Thus $x \Vdash_b a$ means not merely that testing a costs at most b , but that a is true at x and that this truth can be verified within budget b .

Example 9.3 (Medical: Verification costs). For illustration, suppose verified observations carry the following costs:

Test/Observation	Cost	Notes
Fever	\$0	Thermometer
Cough	\$0	Observation
Blood pressure	\$5	Standard equipment
Blood test (WBC)	\$50	Lab work
PCR test	\$100	Molecular
X-ray	\$200	Imaging
CT scan	\$500	Advanced imaging

If patient x really has elevated WBC and that fact costs \$50 to verify, then $x \Vdash_b \text{elevated_WBC}$ for every budget b at least \$50. A negative test result is not represented by a cheap positive incidence; it belongs to the underlying observational vocabulary only if we explicitly include a corresponding generator.

Remark 9.4 (Filtration of forcing). For $b \preceq b'$ we have

$$x \Vdash_b a \implies x \Vdash_{b'} a.$$

Hence the *forcing relations* form a monotone filtration in the budget. No corresponding monotonicity of induced covers or positivity relations is asserted here.

9.2 Budget-indexed covers and positivity: the next problem

At each fixed budget b one may repeat the constructions of the Boolean case:

$$\begin{aligned} a \triangleleft_b U & \quad :\Longleftrightarrow \quad \forall x (x \Vdash_b a \Rightarrow \exists u \in U \ x \Vdash_b u), \\ a \ltimes_b U & \quad :\Longleftrightarrow \quad \exists x (x \Vdash_b a \wedge \Diamond_b x \subseteq U), \quad \Diamond_b x := \{a \in S : x \Vdash_b a\}. \end{aligned}$$

For every fixed b , these are simply the cover and positivity relations induced by the budget-restricted forcing table.

The subtle question is how they vary with b . Although $\Vdash_b$ is monotone in the budget, neither $\triangleleft_b$ nor $\ltimes_b$ inherits a naive monotonicity law automatically. Increasing a budget may introduce new points satisfying the antecedent of a cover, new possible witnesses, and a larger observable profile $\Diamond_b x$; these effects need not point in the same direction. Characterizing conditions under which families $\{\triangleleft_b\}_{b \in B}$ and $\{\ltimes_b\}_{b \in B}$ have useful coherence properties is one of the main tasks deferred to the next article [7].

9.3 Enriched values: an outlook, not a final semantics

A more ambitious route is to replace the Boolean matrix by a value object V carrying both logical and resource information. Schematically one seeks

$$R : X \times S \longrightarrow V$$

together with suitable projections or threshold maps recovering ordinary and budget-indexed forcing. In particular, the Boolean content should remain recoverable from the enrichment rather than being identified with a numerical cost.

If V is equipped with a sufficiently complete residuated monoidal order $(V, \leq, \otimes, I, \multimap)$, the familiar enriched formulas suggest the operators

$$\begin{aligned} (\text{ext}_V \phi)(x) &= \bigvee_{a \in S} \phi(a) \otimes R(x, a), & (\Box_V \psi)(a) &= \bigwedge_{x \in X} (R(x, a) \multimap \psi(x)), \\ (\Diamond_V \psi)(a) &= \bigvee_{x \in X} \psi(x) \otimes R(x, a), & (\text{rest}_V \phi)(x) &= \bigwedge_{a \in S} (R(x, a) \multimap \phi(a)). \end{aligned}$$

Under the usual completeness/residuation hypotheses these have the expected adjoint shape $\text{ext}_V \dashv \Box_V$ and $\Diamond_V \dashv \text{rest}_V$, and suggest an enriched positivity operator $\mathcal{J}_V := \Diamond_V \circ \text{rest}_V$.

We record these formulas only as a bridge to the programme. The precise choice of V , the recovery of the underlying truth relation, the role of thresholds, and the interaction with logical structural rules are not fixed in this survey and will be treated separately [7].

10 Conceptual summary and outlook

Three layers, with different status in this survey.

- (i) forcing and the two adjunctions
- $\Downarrow$
- (ii) cover, positivity, and closedness without complements
- $\Downarrow$
- (iii) resource-aware refinement and logic

Layers (i)–(ii) are developed here. Layer (iii) is the announced research programme.

10.1 Outlook: toward budget-indexed truth

A future resource logic should introduce a new satisfaction judgement, say

$$s \Vdash_b \varphi,$$

where s is an explicitly specified semantic state, φ belongs to a defined syntax, and b is a budget. This judgement is *not* the same object as the incidence relation $x \Vdash_b a$ of Section 9; it would be a recursively defined logical semantics built on top of the resource-refined forcing structure.

For example, if resources must be split between conjunctive subgoals, one might require a relation $\text{Split}(b; b_1, b_2)$ and adopt a clause of the schematic form

$$s \Vdash_b (\varphi \wedge \psi) \iff \exists b_1, b_2 (\text{Split}(b; b_1, b_2) \ \& \ s \Vdash_{b_1} \varphi \ \& \ s \Vdash_{b_2} \psi).$$

Likewise, existential truth should remain witnessed and budget-sensitive. But the syntax, semantic states, budget algebra, structural rules, and exact relation between logical satisfaction and positivity are deliberately left open here. They form part of the next article [7].

A Application to AI engineering

This appendix explains why forcing/positivity/cost provides concrete engineering value for AI systems.

A.1 The AI analogue of forcing

In AI engineering, one evaluates systems against requirements:

- X = executions / traces / agent rollouts
- S = constraints (“tool-call succeeded”, “no disallowed content”, “latency < 500ms”, etc.)
- $x \Vdash a$ = “execution x satisfies constraint a ”

The Boolean forcing table $R \subseteq X \times S$ is therefore an evaluation matrix. The resource-sensitive constructions of Section 9 should be read as possible refinements of this basic evaluation layer.

A.2 Two adjunctions = two engineering operations

(1) $\text{ext} \dashv \Box$: **specification refinement.**

- $\text{ext } U$: which traces satisfy at least one check in U ?
- $\Box E$: which checks are universally valid over traces E ?
- $a \triangleleft U$: whenever a holds, some $b \in U$ holds

This is specification management: what the system guarantees.

(2) $\Diamond \dashv \text{rest}$: **witness finding.**

- $\Diamond D$: which checks are witnessed in trace-set D ?
- $\text{rest } U$: which traces live entirely inside constraint envelope U ?
- $a \ltimes U$: exists a trace satisfying a with profile $\subseteq U$

This is counterexample search and safe operating region identification.

A.3 Positivity = safety envelopes

Engineers want “safe = complement of bad”. This fails because:

- (a) the complement is not observable (unknown unknowns)
- (b) violations are sparse and adversarial
- (c) safety is better defined by positive evidence of robustness

Positivity gives a constructive alternative:

- closed code U with $\mathcal{J}U = U$ is a safety envelope
- $a \ltimes U$ provides witness-based membership

Safety is a robust region, not a complement.

A.4 LLM + PosTop: an illustrative architecture

Current LLM agents can fail by producing claims without witnesses, by combining inconsistent claims, or by spending verification resources poorly. The forcing/cover/positivity picture suggests an engineering discipline in which candidate outputs are checked against explicit observations, witnesses, and budgets.

At this point, however, a typing distinction is essential. The present survey has defined cover and positivity for generators $a \in S$ and subsets $U \subseteq S$; it has *not* identified arbitrary LLM hypotheses with generators. A concrete system must therefore provide an encoding of hypotheses into the formal structure – for example a map $\iota : H \rightarrow S$ or a richer syntax built over S – together with a representation of the current observation state. Only after that choice is made do expressions involving cover or positivity become well typed.

One possible implementation pattern is then:

1. the LLM proposes candidate hypotheses H ;
2. an explicit encoding maps each candidate into the PosTop structure;
3. cover checks test the chosen direction of semantic consequence, while positivity checks demand a witness;
4. a resource layer proposes an observation or test that best discriminates among the remaining candidates;
5. the process repeats until the task is resolved or the budget is exhausted.

This is an architectural *schema*, not a theorem of the present paper. In particular, the direction of the cover check depends on how an observation state and a hypothesis are represented; the informal expression “ $h \triangleleft \text{observed}$ ” used in earlier drafts should not be read as defined notation. A fully typed account of hypotheses, observations, resource choice, and agent traces belongs to the subsequent development [7].

Comparison with knowledge graphs. The comparison with knowledge graphs is best understood as a difference of emphasis, not as an impossibility claim. A bare knowledge graph primarily records entities and relations; entailment procedures, provenance, costs, and verification policies may certainly be layered on top. Positive Topology contributes a native language for cover, positivity, witness preservation, and – in the proposed extension – feasible refinement.

A knowledge graph is naturally a representation of structured facts. The PosTop programme aims to add a grammar of refinement: what follows, what remains witnessable, and eventually what can be verified within available resources.

B Software and Reproducibility

The LaTeX file you are reading lives in the same repository as the reference implementation of the forcing matrix, adjunctions, and notebooks. The codebase is organized as follows:

- `src/postop/core.py` implements the data structures discussed in Sections 4–6, with the forcing relation encoded as a Python dictionary of point $\rightarrow$ observable sets, plus the cost-aware variants in `src/postop/costs.py`.
- `src/postop/operators.py` and `src/postop/llm.py` provide tracing/explainability helpers that surface the cover witnesses/counterexamples required by the diagnostic examples.
- `examples/` mirrors the running medical and guardrail scenarios; `tests/` exercises the adjunction identities via `pytest`.
- `docs/REPRODUCIBILITY.md` gives a copy/paste runbook matching the commands below.

The latest sources live at <https://github.com/mirco-mannucci/postop>; we snapshot paper+code together via the git tag `v0.1.0` (and subsequent releases) so the commands below always map to a reproducible commit/DOI pair.

B.1 Environment setup

We keep the environment lightweight (pure Python + `pytest`). From the repository root:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -e .[dev]
```

B.2 Running the examples

The forcing table is serialized as ordinary Python dictionaries and can be inspected directly. To reproduce the medical walkthrough and the illustrative guardrail example discussed in this survey:

```
python examples/medical.py
python examples/llm_guardrail.py
```

Each script prints the `ext/□/◇/rest` results, the compatibility witnesses, and the guardrail counterexamples that correspond to the formal statements in the main text.

B.3 Notebooks and tests

The ActionList roadmap adds four short notebooks under `notebooks/`; they can be executed headlessly via:

```
jupyter nbconvert --to notebook --execute \  
  notebooks/01_basics_ext_int_hit_sel_j.ipynb
```

Swap in the remaining notebook filenames (medical diagnosis, costs, guardrails) to reproduce the figures in the extended version of the paper.

Finally, run the test suite to check the adjunction/proof obligations:

```
pytest -q
```

These tests import the installed package so they are venv-friendly and catch regressions both in the mathematical invariants (idempotence/contractivity) and in the explainability helpers referenced by the notebooks.

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